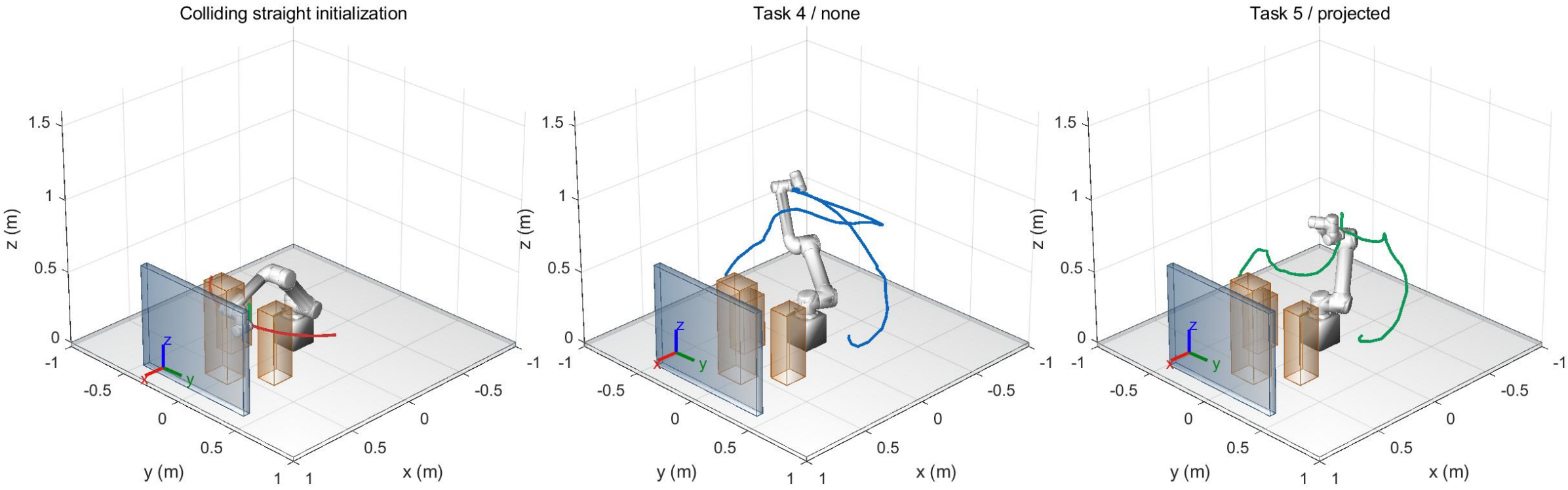
Task 4: a mounted UR5 workcell



92 / 181

initial states collide

Same endpoints and straight initialization for both tasks

Task 5: upright tool direction

Tool y aligns with world +z

Yaw remains free

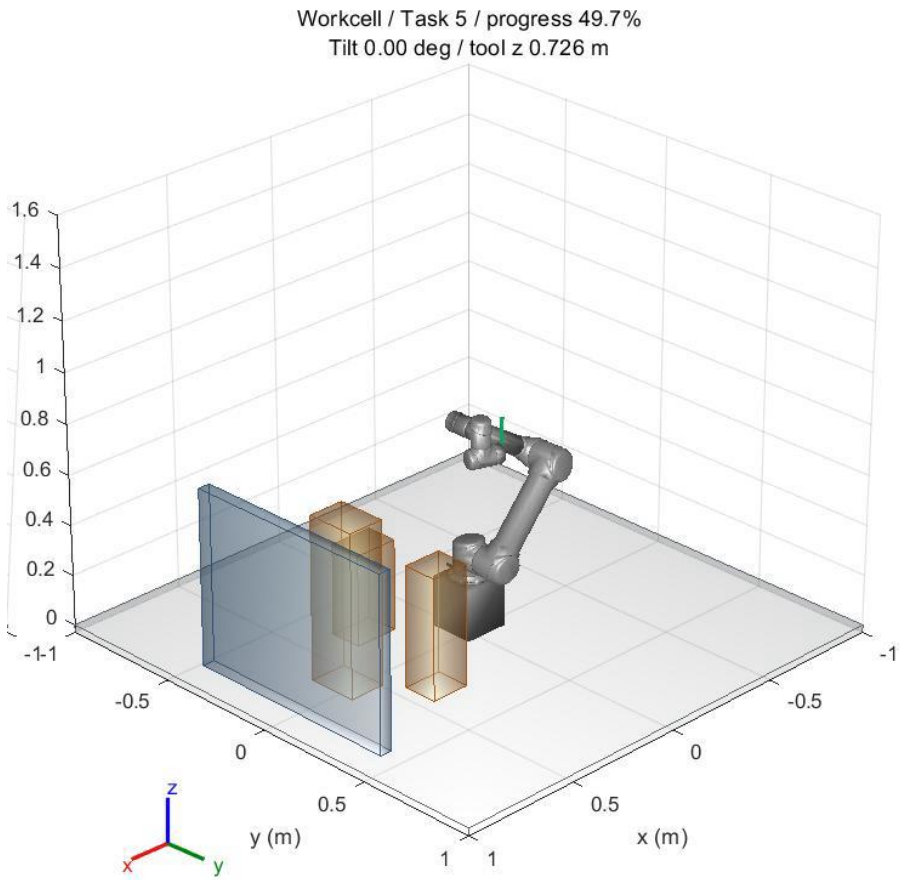
Direction penalty, lambda = 1000

Damped Jacobian projection

Self-collision cost and rejection

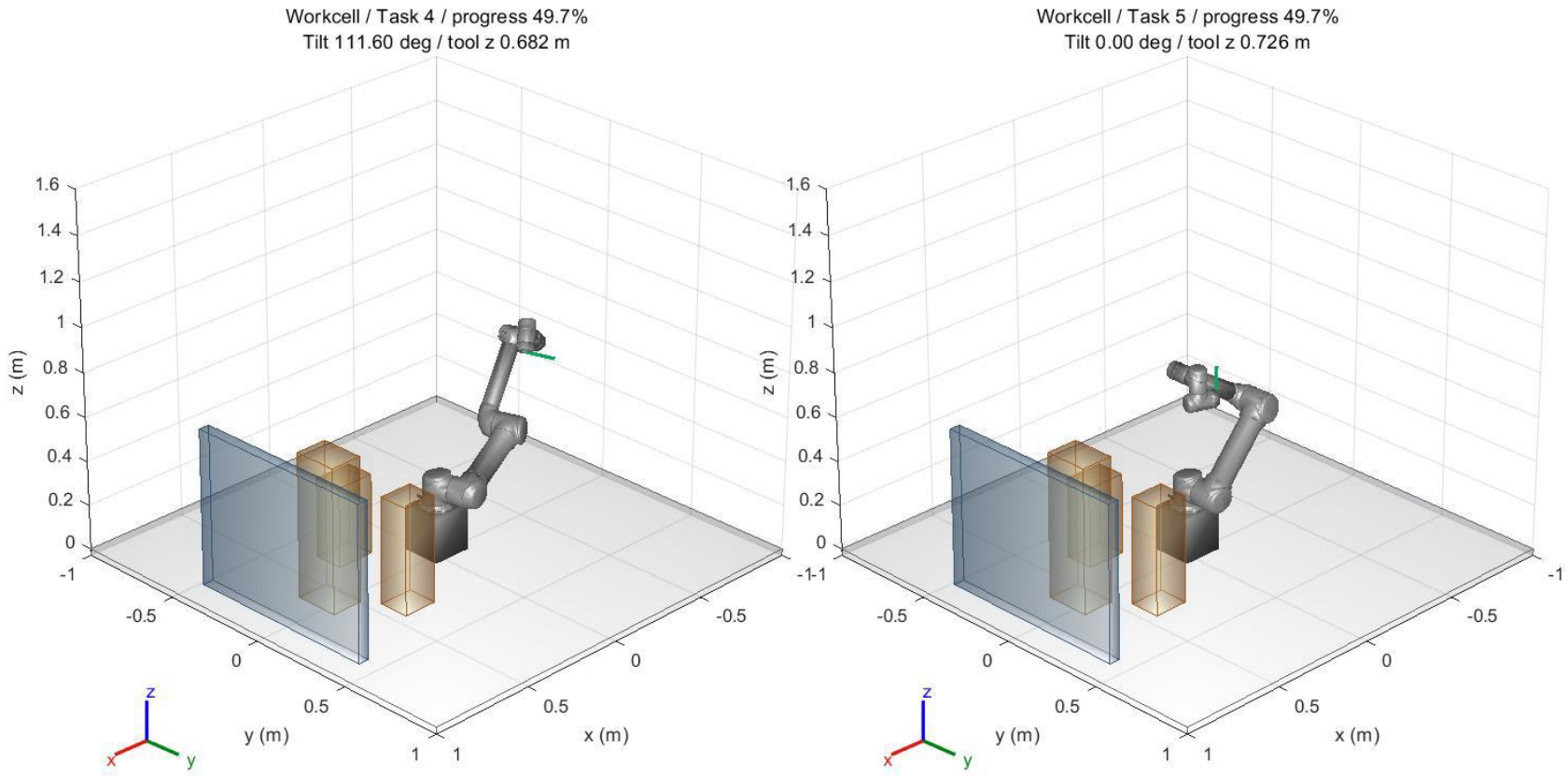
Maximum accepted tilt: 5 degrees

Conservative clearance: 35 mm



Click video to play / 10 s

Same problem, different tool orientation



Demo maximum tilt: 120.29 degrees (Task 4) / 0.930 degrees (Task 5)

Seed 5300. Normalized path progress, without an execution-time model.

Held-out results and contribution

Condition	Success	Median time (s)	Worst tilt (deg)	Min proxy gap (mm)
Task 4	20/20	30.42	169.591	35.41
Task 5	20/20	68.19	0.997	35.23

20 paired seeds. Every successful path also passes the 0.00625 rad check.

My contribution: scene, upright projection, self-collision handling, paired experiments and independent verification.

V4/V5 extensions in backup slides. Discrete geometry checks; no payload dynamics.

Project overview: Tasks 1 to 5

Task	What changes	Evidence
1	Complete KINOVA STOMP	Original scene, collision-free animation
2	Replace KINOVA with UR5	UR5 planning under the same framework
3	Replace planning FK with PoE	All-body accuracy and timing comparison
4	Design a harder UR5 workcell	Colliding initialization and planned route
5	Keep the tool direction upright	Same-problem constrained comparison

Reuse the team baseline. Member C adds Tasks 4–5 and the later ablations.

STOMP iteration in this implementation

Stage	Operation	Relevant function
Sample	Correlated, antithetic perturbations	stompSamples
Evaluate	World / self / direction / smoothness	stompTrajCost
Weight	Lower local cost gets higher probability	stompUpdateProb
Update	Weighted actual noise, then smoothing	stompDTheta / stompUpdateTheta
Accept	Limits, projection and feasibility gates	memberC.plan / validatePath

Optimize internal waypoints. Keep the endpoints fixed. No global-optimality guarantee.

PoE forward kinematics

S = [a; -a cross p]

T(q) = exp([S1]q1) ... exp([Sn]qn) M

Check	Saved Task 3 baseline
All-body position error	6.65e-16 m over 200 configurations
Task 2 vs Task 3 path difference	0 rad in the fixed-seed run
Same all-body workload speed ratio	22.85x (original Linux machine)

Cache screws and prefixes once. Timing depends on machine and workload.

Direction constraint and projection

$y(q) = R(q)\, e_y$

$\text{Tilt} = \arccos(\text{clamp}(y(q)^T e_z, -1, 1))$

$J_y(:,i) = \omega_i \times y$

$\Delta q = J_y^T (J_y J_y^T + \mu^2 I)^{-1} (e_z - y)$

Projection setting	Value
Damping / internal tolerance	0.001 / 1 degree
Steps / component update cap	20 / 0.15 rad
Final dense tilt tolerance	5 degrees

What changed across the iterations

Version	Main change	Observed result
V1	Penalty + direction projection	Projected Task 5: 0/20, self collisions
V2	Self-collision handling, feasible incumbent	Task 5: 20/20 across two cases
V3	Mounted table workcell	Task 4 and Task 5: 20/20 each
V4	Feasible refinement + tangent sampling	Four arms: 10/10 each
V5	Energy control + adaptive ablation	71/75, four margin failures

Different scenes and seeds prevent causal cross-version speed claims.

V4: refinement and tangent sampling

Task 5 arm	Success	Median time (s)	Median final Q
baseline	10/10	49.96	10.15
anytime	10/10	94.94	9.88
tangent	10/10	39.88	9.76
combined	10/10	89.12	9.39

Combined vs baseline: median paired Q reduction 15.36% (n=10)

7 wins, 3 losses. Median paired extra time: 37.77 s.

Q = joint length + 5 x smoothness + soft 45 mm clearance preference

V5: three cases, five sampling strategies

Task 5 arm	Workcell	Endpoints	Layout	Time (s)	Final Q
ambient	4/5	5/5	5/5	103.57	8.64
energy_control	5/5	5/5	5/5	92.94	8.78
fixed50	5/5	4/5	5/5	151.12	9.04
fixed75	5/5	5/5	5/5	90.80	9.03
adaptive	4/5	4/5	5/5	98.86	8.39

Fixed75: 15/15. Adaptive: 13/15. Keep adaptive experimental.

Time uses all runs. Q uses successes. Adaptive vs fixed75: 4.60% paired Q reduction, n=13.

Four V5 failures miss the clearance gate

Case / arm / seed	Proxy gap (mm)	Tilt (deg)
workcell / ambient / 5800	30.13	0.152
workcell / adaptive / 5801	30.42	0.511
endpoints / fixed50 / 5800	34.22	0.083
endpoints / adaptive / 5801	33.53	0.858

Required conservative gap: 35 mm. All four tilts are below 1 degree.

Soft-cost feedback can disagree with dense feasibility.

This is a plausible mechanism, without causal proof from these runs.

Energy control isolates a sampling effect

Sampler	Pre RMS (rad)	Post RMS (rad)	Corrections / point
ambient	0.3761	0.3417	2.6578
energy control	0.3371	0.3051	2.3019
fixed75	0.3371	0.3273	0.7119

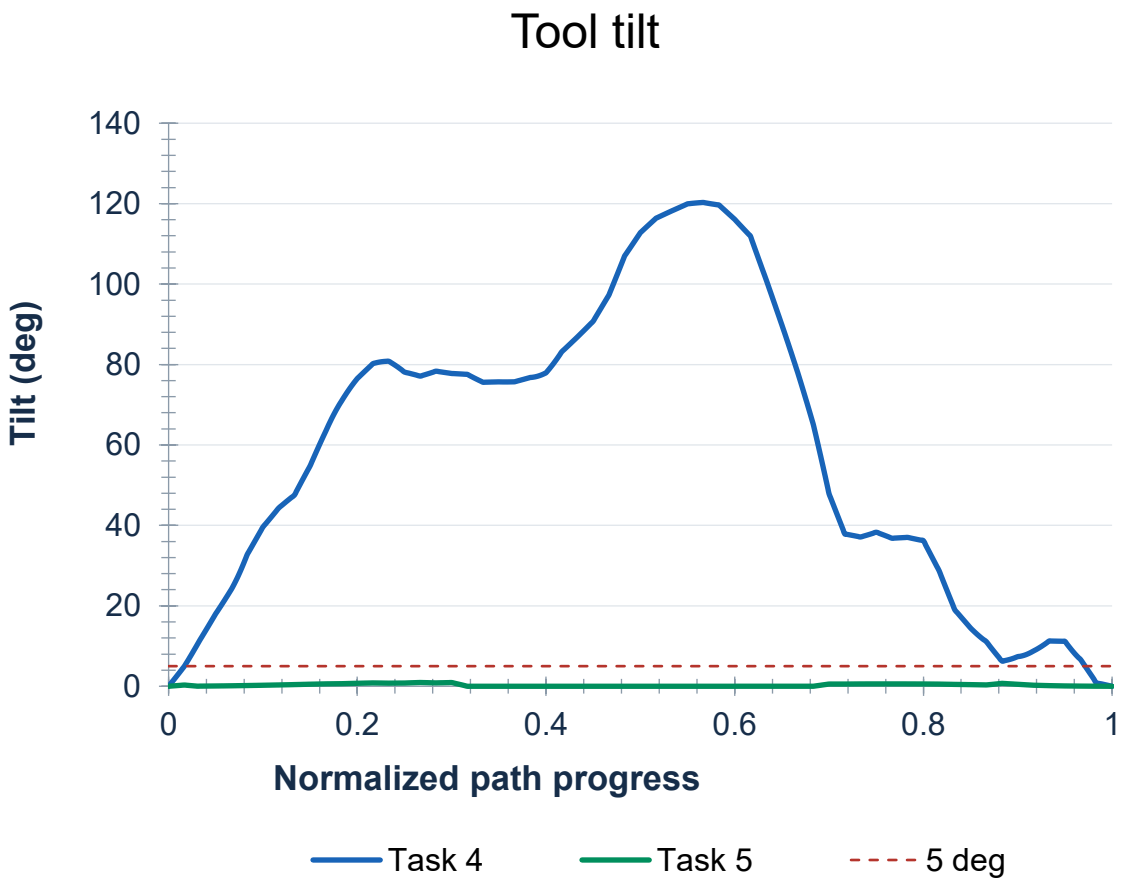
Fixed75 vs control: median correction reduction 68.04%

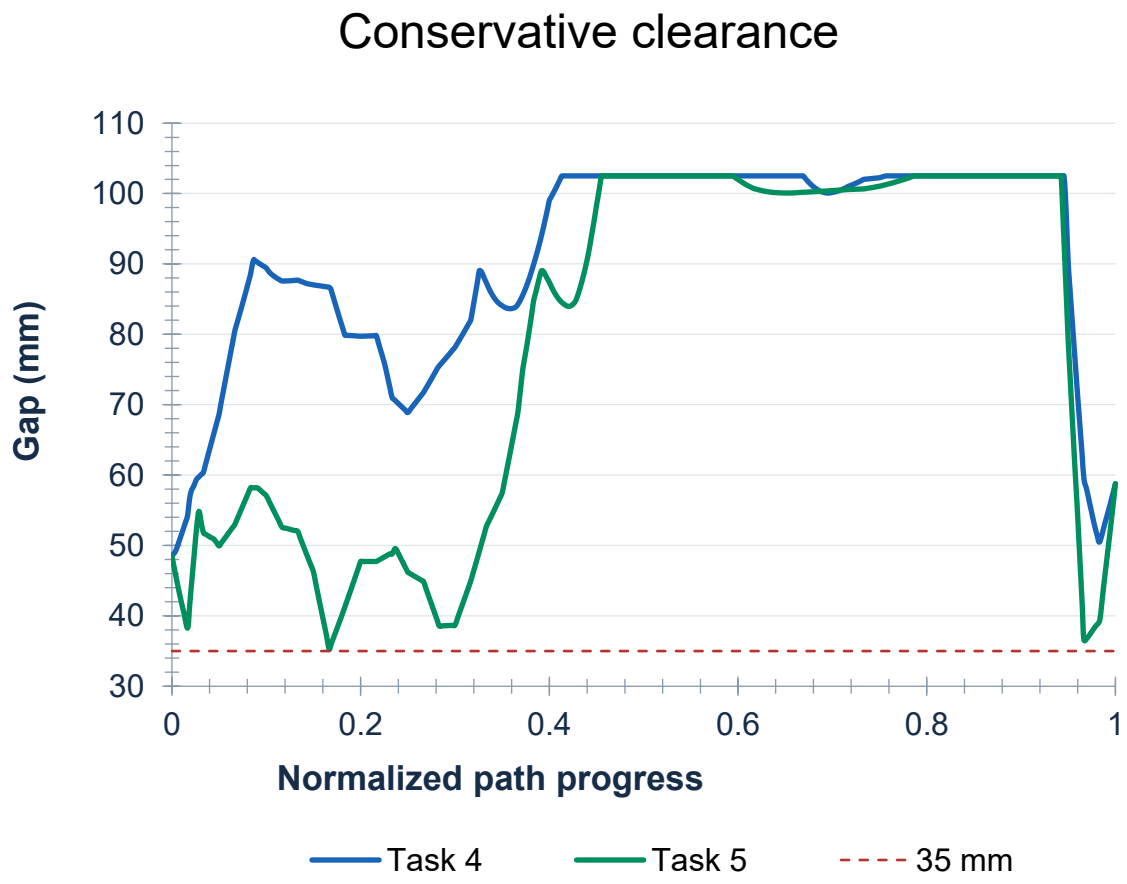
Match scalar input norm before nonlinear projection.

30 paired batches reuse five seeds over six central paths.

Formal planning shows no consistent Q advantage over the energy control.

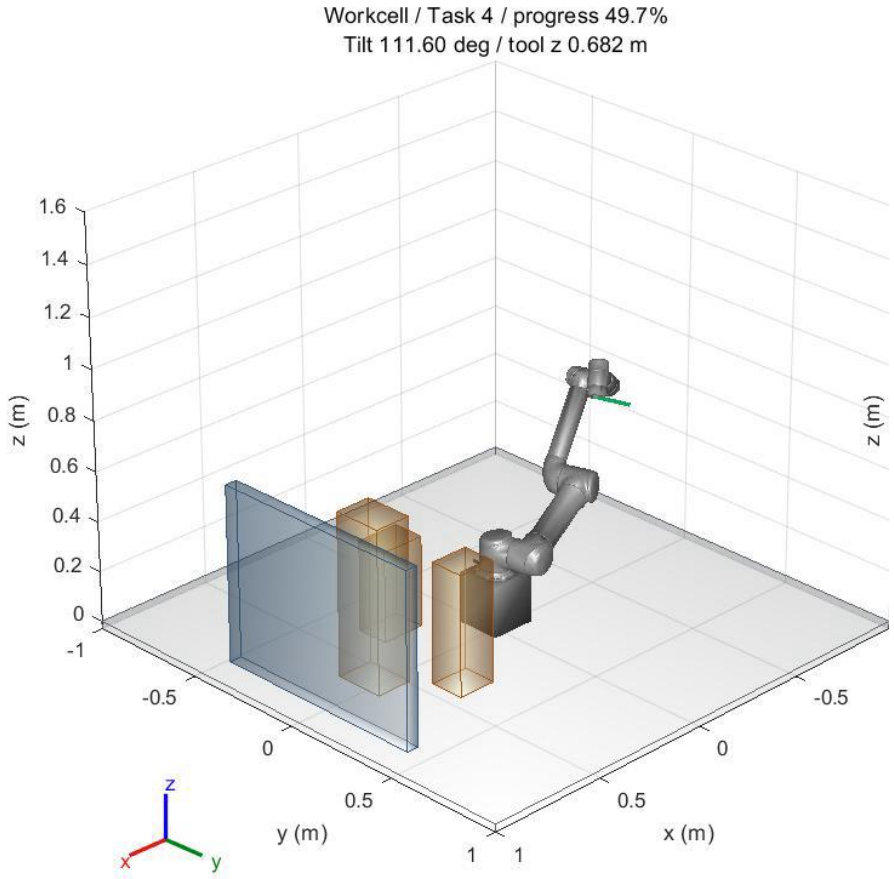
Demo profiles from the saved dense paths





Development seed 5300. Native charts retain the full saved CSV data.

Task 4 standalone motion



World and self-collision checks pass.

Tool orientation remains free.

Demo maximum tilt: 120.29 degrees.

Click video / 10 s

Same seed, endpoints and workcell as the main comparison.

Evidence, contribution and remaining limits

Evidence	Scope
V3 formal / demo replay	40 paths, 804 checkpoints / 2 paths, 20 checkpoints
V4 formal replay	40 paths, 1365 checkpoints
V5 formal replay	75 paths, 3005 checkpoints
Finer geometric check	Successful final paths at 0.00625 rad joint increments

Member C: workcell, upright and self-collision extensions, controlled experiments, independent verification and delivery.

Discrete checks only. No payload geometry, dynamics, hardware execution or global guarantee.